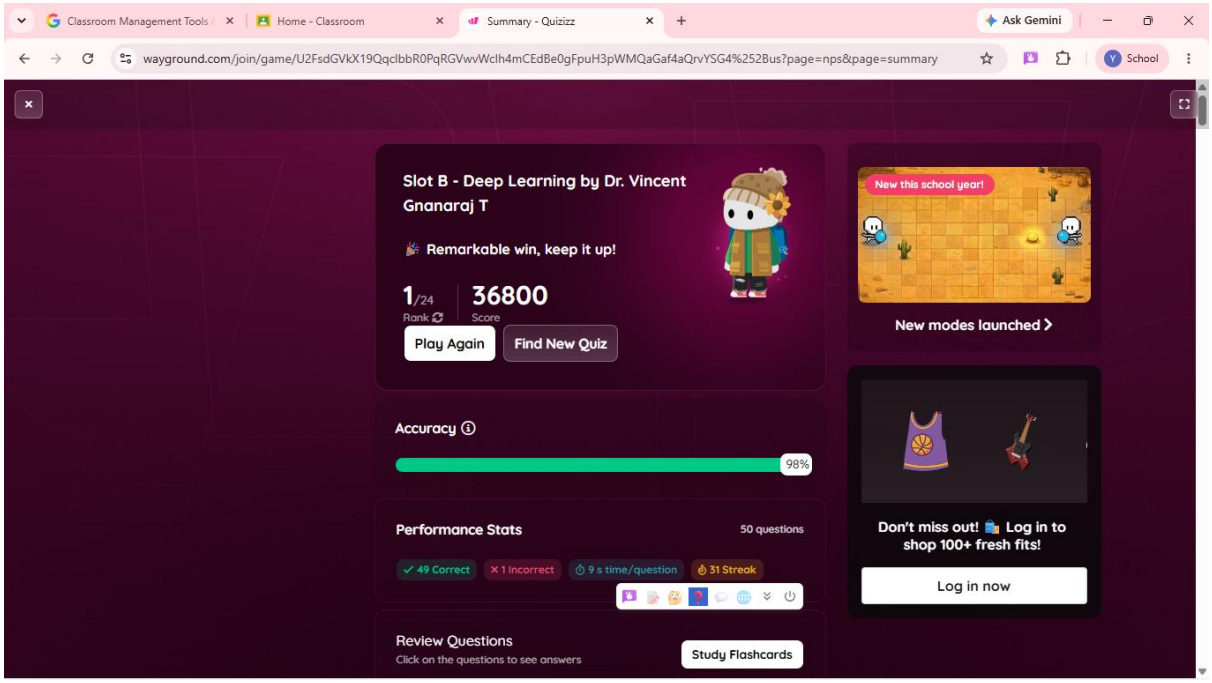
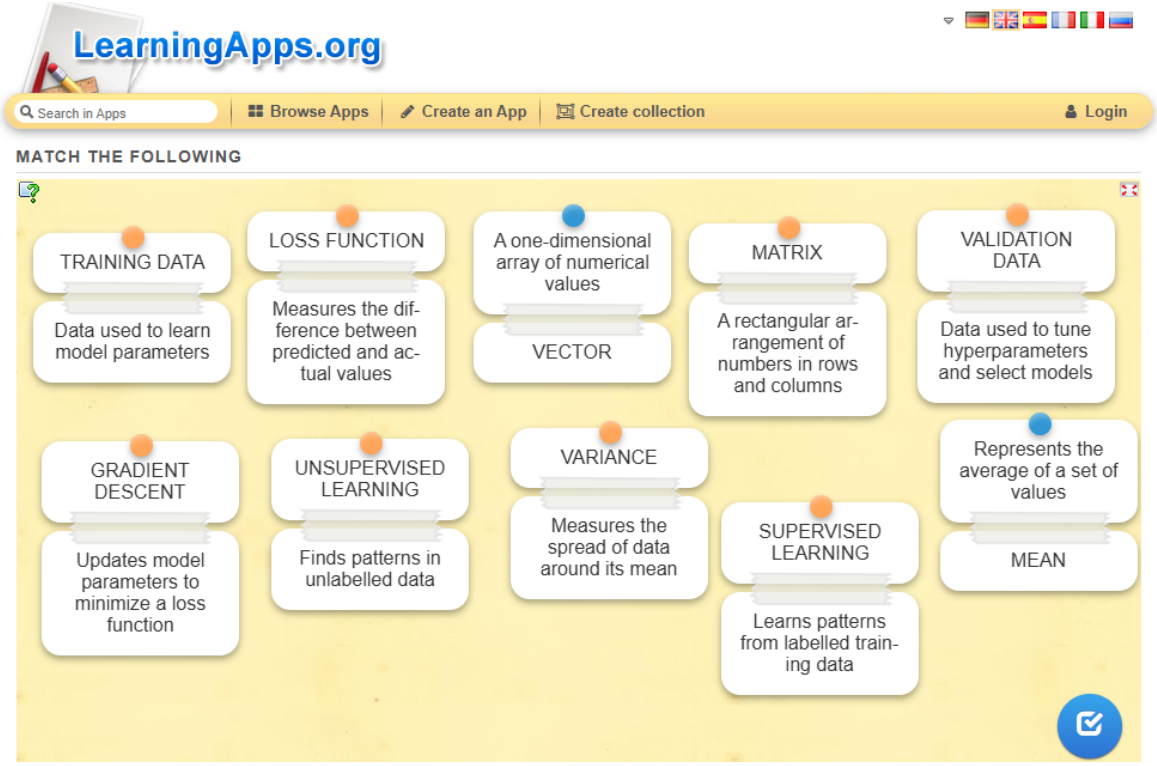
CSA4701 – DEEP LEARNING NAME : YUGITHA B

CO1 – ASSESSMENTS REG NO : 192424102





In machine learning, mathematical concepts are essential for representing data, developing algorithms, and building predictive models. A Vector is a one-dimensional array of numerical values, while a Matrix is a rectangular arrangement of values organized into rows and columns. Matrix operations such as addition, multiplication, and transposition are widely used in deep learning computations. The Mean represents the average value of a dataset, whereas the Variance measures how widely the values are distributed around their mean. The Standard deviation measures the square root of the variance and expresses data variability in the original unit of measurement. In probability theory, the Probability represents the likelihood of an event occurring, with values ranging from zero to one. A covariance describes the relationship between two variables and indicates how they vary together.

Machine learning algorithms use data to identify patterns and make predictions. In Supervised learning, models are trained using labelled datasets containing input features and their corresponding target outputs. In contrast, Unsupervised learning identifies hidden patterns, groups, or structures in datasets without predefined labels. Reinforcement learning involves an agent interacting with an environment and learning suitable actions through rewards and penalties. The dataset used to learn model parameters is called the Training dataset, while the Validation dataset helps tune hyperparameters and select suitable models. The Test dataset is used to evaluate the final model's performance on previously unseen data.

During model training, a Loss function measures the difference between predicted outputs and actual target values. The Gradient descent algorithm updates model parameters iteratively to minimize this loss. The Learning rate controls the size of each parameter update during optimization. If this value is too large, training may become unstable; if it is too small, convergence may be slow. A Parameter is a parameter learned automatically during training, whereas a Hyperparameter is configured before training begins. Examples of hyperparameters include learning rate, batch size, and number of training epochs.

A machine learning model may experience Overfitting when it performs well on training data but poorly on unseen data. Conversely, Underfitting occurs when a model is too simple to capture important patterns in the data. The balance between these problems is related to model Gradient, which describes its ability to represent different functions and relationships. Regularization is a technique used to reduce overfitting by penalizing overly complex models or large parameter values. In neural networks, an Activation function introduces non-linearity, enabling the network to learn complex relationships. Finally, Evaluation refers to the process of assessing model performance using appropriate evaluation metrics, while Generalization